

AI-Assisted Discovery

Lecture 1

Stan Fedotov
Nebius Academy



Nebius: The Vertical AI Company

GPU cloud

Top-notch NVIDIA GPUs:
H100, H200, B200, GB200, GB300

Global AI infrastructure provider

7+ data centers around the globe

NEBIUS ACADEMY

Nebius Academy is the education and research arm of Nebius.

Token Factory

Inference and fine-tuning

AI R&D

In-house research lab

Open applied research

Papers, code, data



NEBIUS ACADEMY

A global community network for applied AI across Israel, the UK, the US, and Europe.

Scope & Scale:

a core community of ~3,000
top AI and tech
professionals

Who's Inside:

AI/ML engineers, software
developers, researchers,
product and data leaders,
university partners

What We Do:

Open
courses, expert webinars,
Discord groups, partner
workshops, and in-person
meetups



NEBIUS ACADEMY

Research collaborations in AI-assisted Math Discovery



Caltech

AI for Mathematical Sciences seminar series (AIMS)

AI and the formalisation of mathematics

2 PM, 9 MAR 2026

Kevin Buzzard opens AIMS with his views on what a new era of formalised maths, Lean and AI—verified proofs mean for the future of research.

Event information

This event, part of our AI for Mathematical Sciences series, takes place at 2 pm on Monday 9 March at the London Institute for Mathematical Sciences, on the second floor of the Royal Institution. AIMS is sponsored by Nebius. To register for the series please fill out the [online form](#).

Speaker



Kevin Buzzard is a professor of pure mathematics at Imperial College London. He specialises in arithmetic geometry, number theory and the Langlands programme and leads work on formalising mathematics with computer proof assistants, including projects in the Lean theorem prover.



<https://lims.ac.uk/event/ai-and-the-formalisation-of-mathematics/>

Join our Discord!



<https://discord.gg/9Bw5e3zkjk>

About me

Stan Fedotov

Before 2013 – PhD in Maths

Did Representation Theory & Invariant Theory
(representations of quivers)

Since 2013 – managing and leading educational
projects in ML, later in AI

Now – AI Program Lead at Nebius Academy in London

Since 2025 – returned to Maths research as an
experimentalist (wrote a paper with Evgeny Feygin in
December 2025)



What AI can do for Maths

1. Proving theorems
2. Verifying proofs (not really AI)
3. Discovering new theorems
- ~~4. Creating new theories (!?)~~

Course plan

1. LLMs and agents as research assistants
2. ML: approximation as a discovery tool
3. A closer look at LLMs and agents
4. Accelerating search in large spaces

Today: theorem proving and verification

1. What AI is
2. LLMs as provers and assistants + why they fail
3. Formal verification and proving + a short intro to Lean
+ proving-and-verification loop

What is AI?

Data Analysis as a driver of discovery

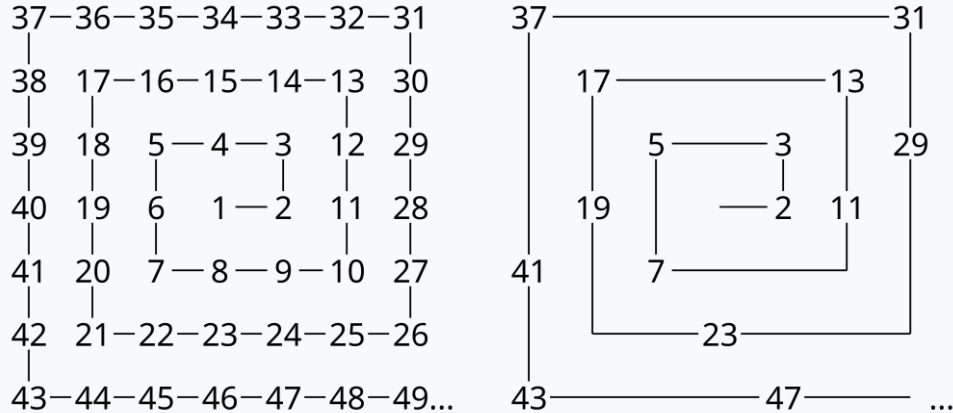
Prime numbers: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47

Data Analysis as a driver of discovery

Prime numbers: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47

Data Analysis as a driver of discovery

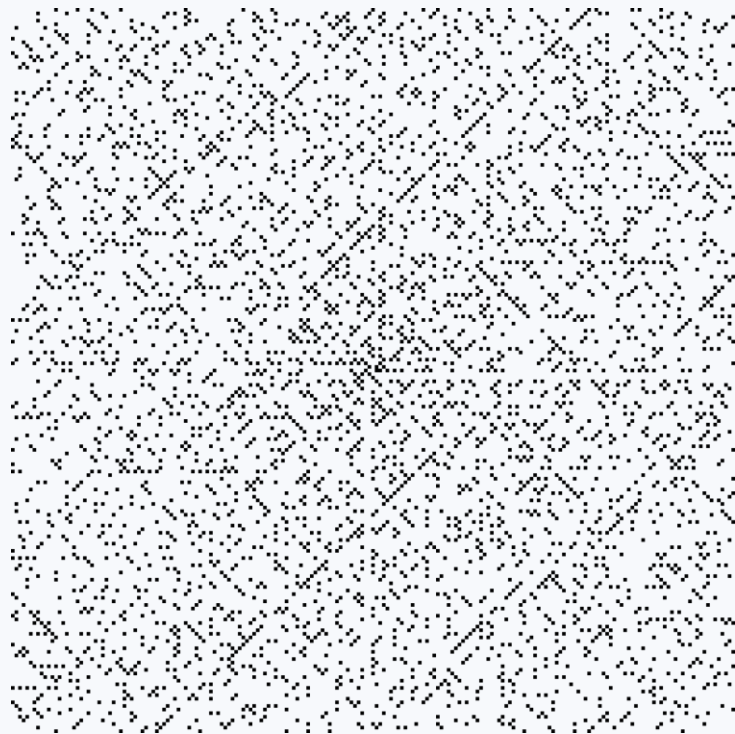
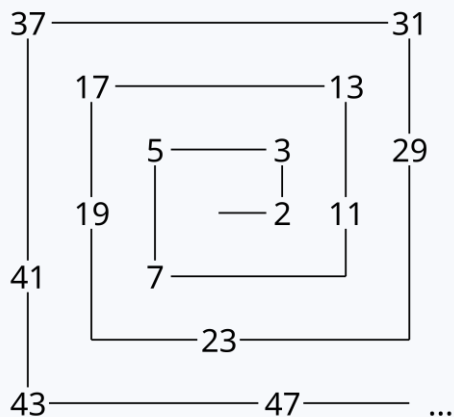
Example: Ulam spiral



Data Analysis as a driver of discovery

Example: Ulam spiral

37	36	35	34	33	32	31
38	17	16	15	14	13	30
39	18	5	4	3	12	29
40	19	6	1	2	11	28
41	20	7	8	9	10	27
42	21	22	23	24	25	26
43	44	45	46	47	48	49...



Lines: $4x^2 + bx + c$

Data Analysis as a driver of discovery

Example: Ulam spiral

Hardy–Littlewood Conjecture (F). *If a, b, c are relatively prime integers, a is positive, $a + b$ and c are not both even, and $b^2 - 4ac$ is not a perfect square, then there are infinitely many primes of the form $f(t) = at^2 + bt + c$. The number of such primes at most x is asymptotic to*

$$\epsilon \prod_{\substack{p \geq 3 \\ p \mid \gcd(a,b)}} \frac{p}{p-1} \prod_{\substack{p \geq 3 \\ p \nmid a}} \left(1 - \frac{(\Delta/p)}{p-1}\right) \text{Li}(x), \quad (6.5.1)$$

in which

$$\epsilon = \begin{cases} \frac{1}{2} & \text{if } 2 \nmid (a+b), \\ 1 & \text{otherwise.} \end{cases}$$

Lines: $4x^2 + bx + c$

<https://www1.cmc.edu/pages/faculty/lenny/papers/bateman-horn.pdf>

The Birch and Swinnerton-Dyer (BSD) conjecture

Curve $y^2 = x^3 + ax + b$; $a_p(E) = p + 1 - \#E(\mathbb{F}_p)$. L-series:

$$L(C, s) := \prod_{p \nmid 2\Delta} (1 - a_p p^{-s} + p^{1-2s})^{-1}.$$

Conjecture (Birch and Swinnerton-Dyer). *The Taylor expansion of $L(C, s)$ at $s = 1$ has the form*

$$L(C, s) = c(s - 1)^r + \text{higher order terms}$$

with $c \neq 0$ and $r = \text{rank}(C(\mathbb{Q}))$.

Observed in 1960s.

<https://www.claymath.org/wp-content/uploads/2022/05/birchswin.pdf>

Machine Learning: solving the task of prediction (functional approximation)

The task: Obtaining an approximation for the smallest root of

$$x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0 = 0$$

It's a function

$$(a_4, a_3, a_2, a_1, a_0) \rightarrow \text{smallest root}$$

ML way of doing this:

Step 1: Get data

Training data

$(p = 0, q = 0) \rightarrow 0$
 $(p = 0, q = -1) \rightarrow -1$
 $(p = 2, q = 1) \rightarrow -1$
...

Predicted target:

The smallest root
of $x^2 + px + q = 0$

ML way of doing this:

Step 2: Choose a model class

Model class

$$\sum_{k,l=0}^M w_{kl} p^k q^l$$

Model parameters

Training data

$$\begin{aligned}(p = 0, q = 0) &\rightarrow 0 \\(p = 0, q = -1) &\rightarrow -1 \\(p = 2, q = 1) &\rightarrow -1 \\&\dots\end{aligned}$$

Predicted target:

The smallest root
of $x^2 + px + q = 0$

ML way of doing this:

Step 2: Choose a model class

Model class

$$\sum_{k,l=0}^M w_{kl} \sin(kp) \cos(lp)$$

Model parameters

Training data

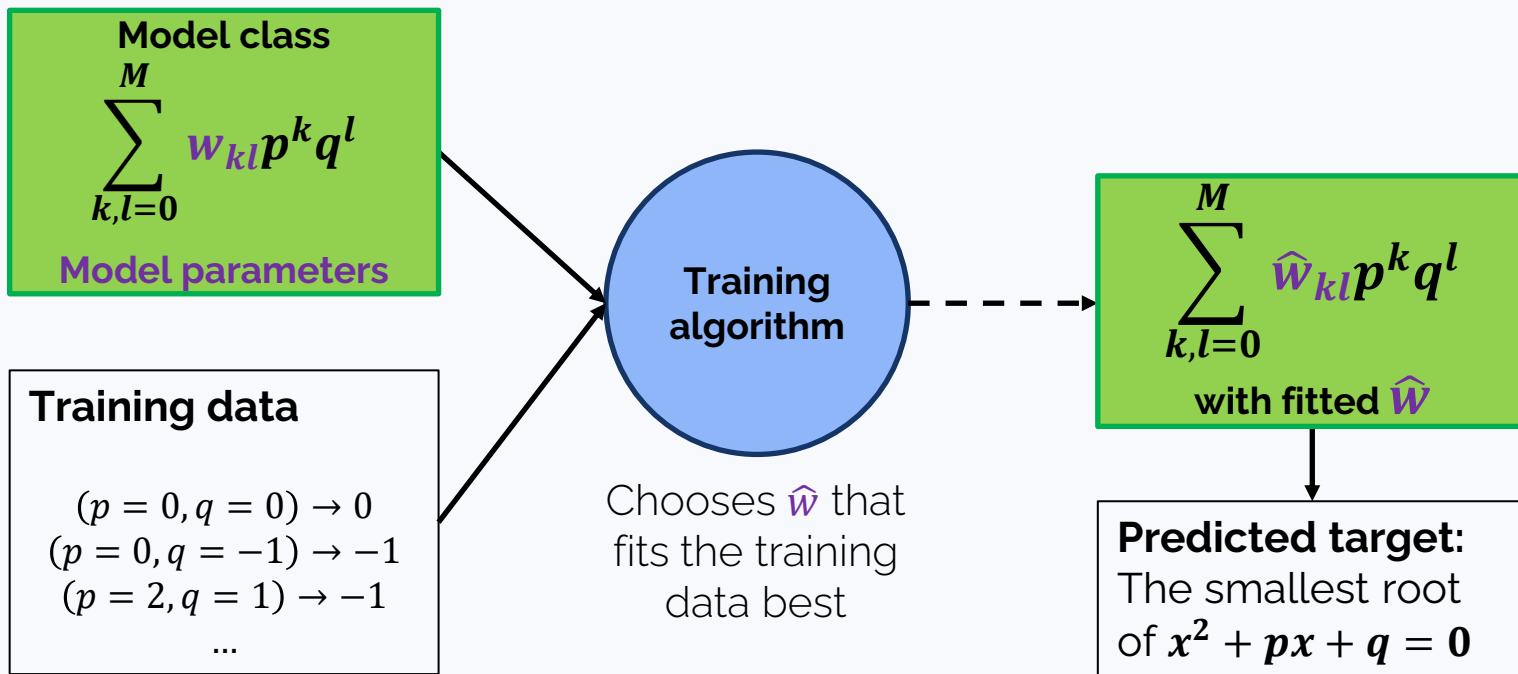
$$\begin{aligned}(p = 0, q = 0) &\rightarrow 0 \\(p = 0, q = -1) &\rightarrow -1 \\(p = 2, q = 1) &\rightarrow -1 \\&\dots\end{aligned}$$

Predicted target:

The smallest root
of $x^2 + px + q = 0$

ML way of doing this:

Step 3: Choose a training algorithm

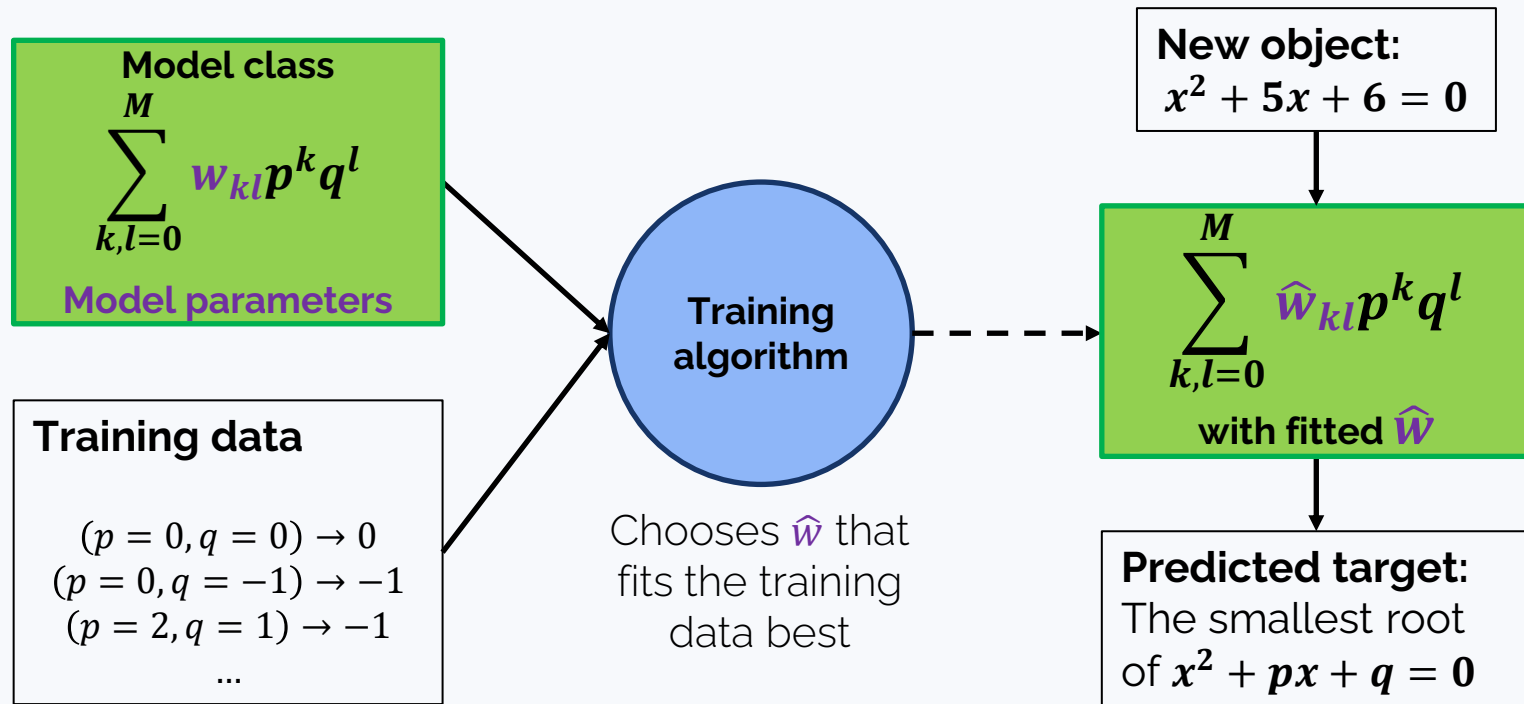


Supervised learning

Trained on
(data, target) pairs

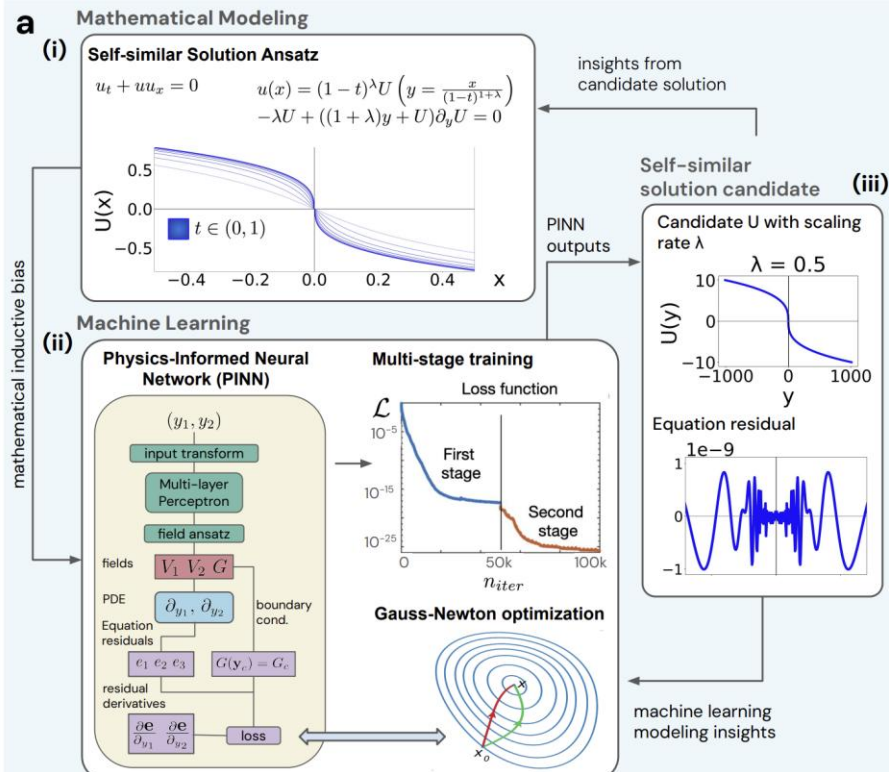
ML way of doing this:

Step 4: Enjoy the outcome (or not)

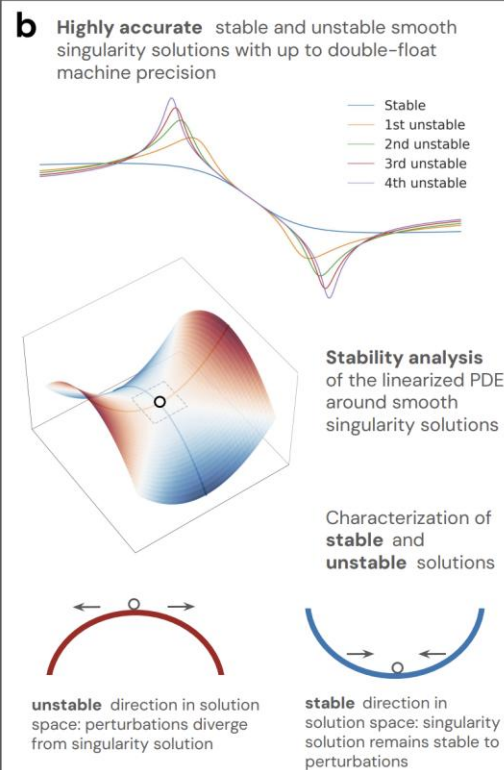


A solution obtained through Machine Learning

High Precision Self-Similar Solution Discovery



Solution Analysis



Discovery of Unstable Singularities

<https://arxiv.org/abs/2509.14185>

This ML paradigm has its limits

Complex tasks require

1. Larger models
2. (Much) more data

Imagine:

- Image segmentation
- Text summarization



Idea: pre-train simple, then fine tune

A pre-trained model

Fine
tuning

Consumer model

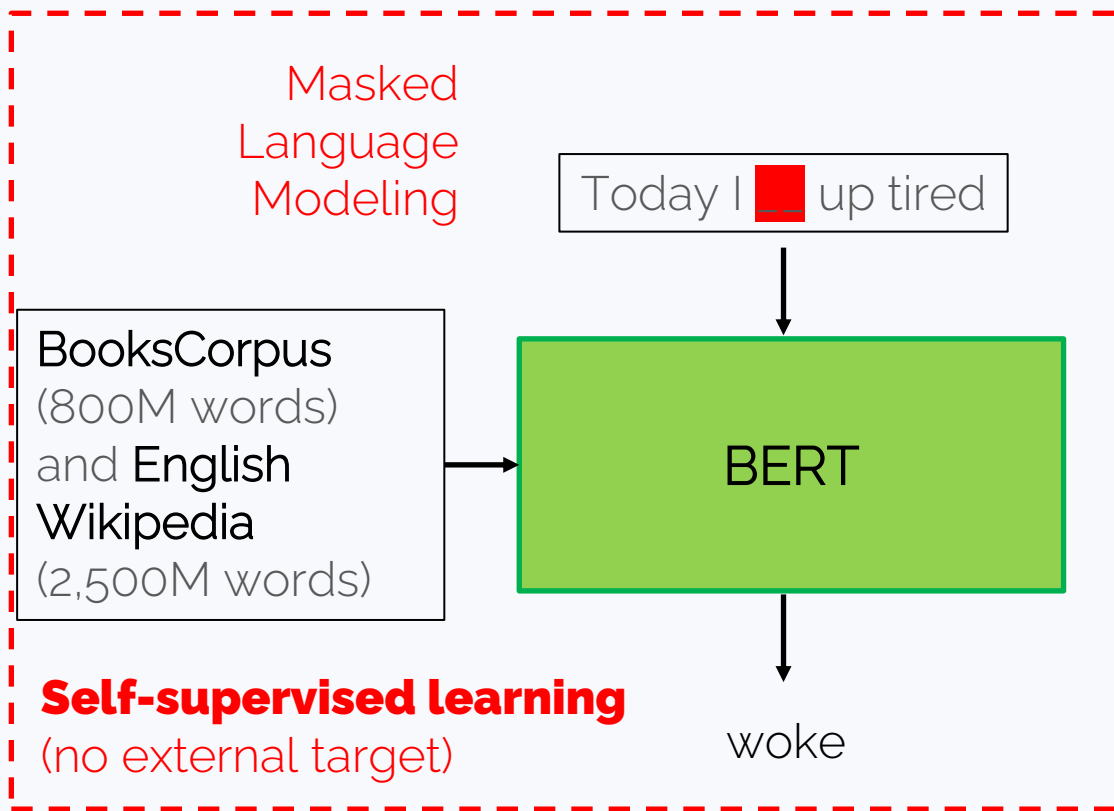
Trained

- For a simple task
- On abundant data
- By a rich corporation or a cool research lab

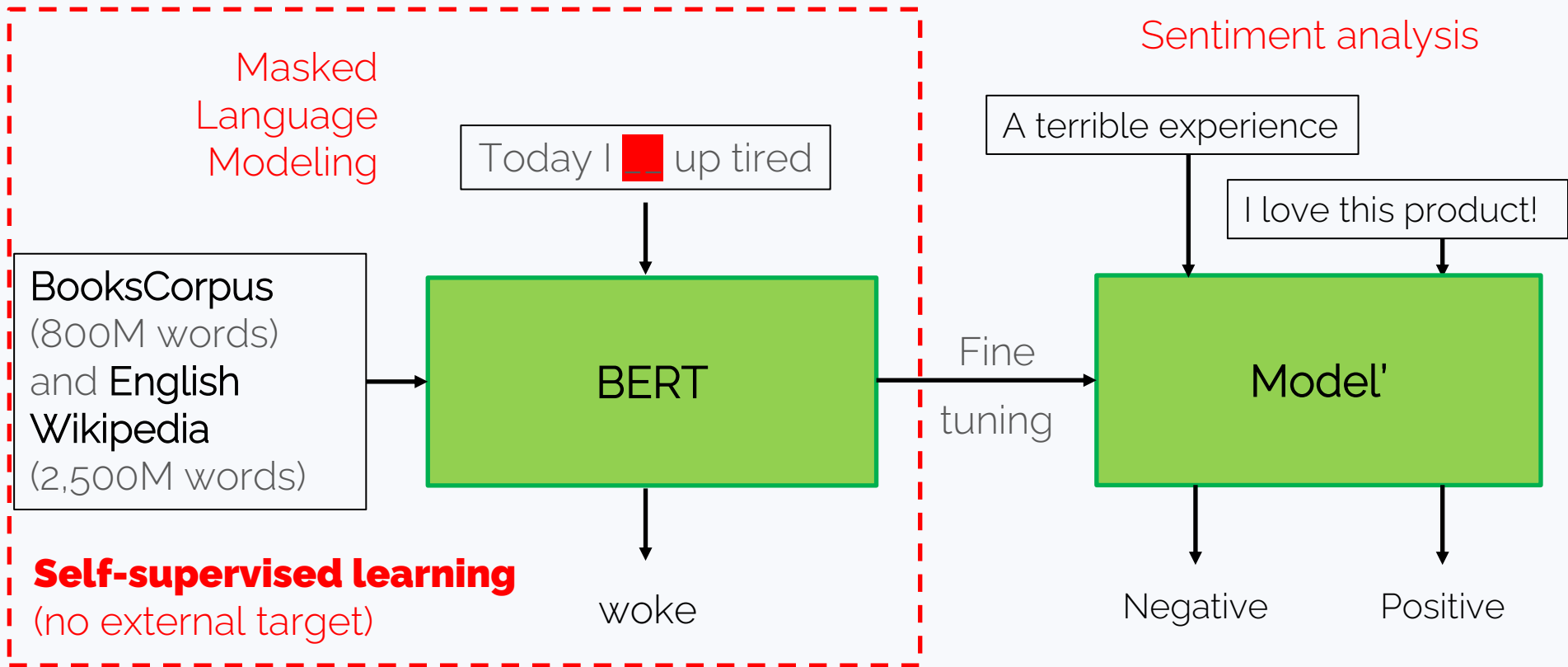
Fine tuned

- For a more niche task
- On scarce data

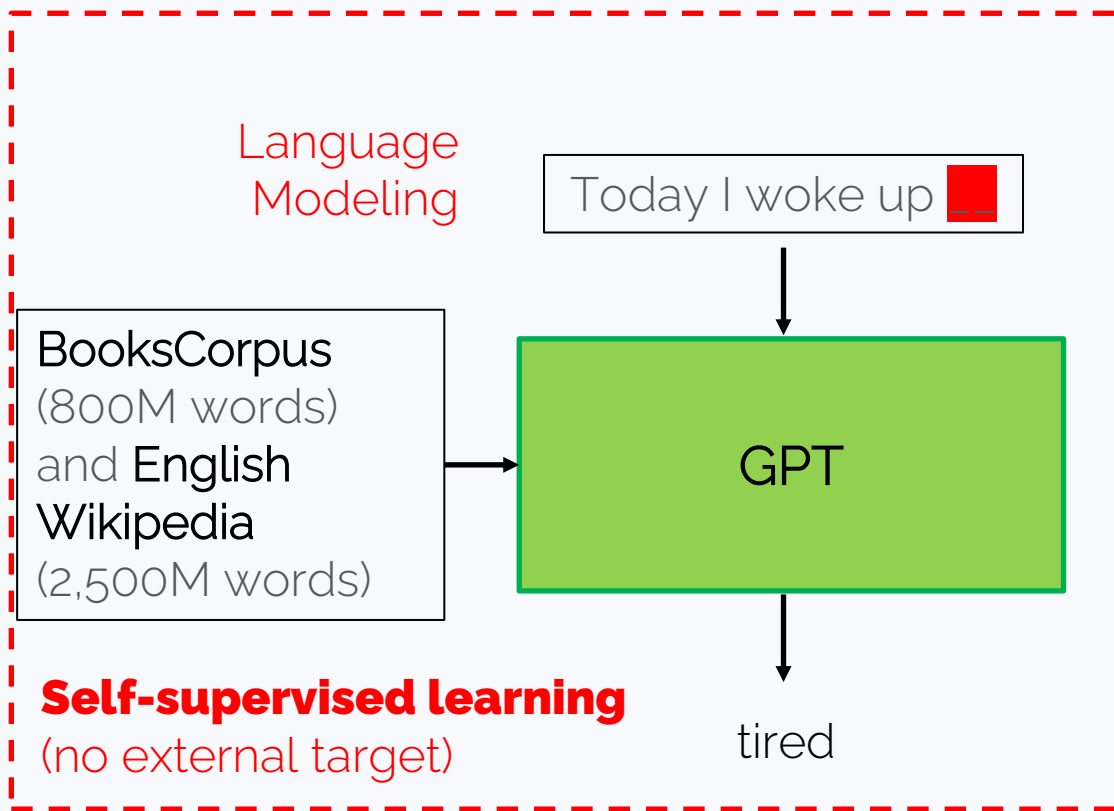
Mid-2010s-2022: the era of large pre-trained models



Mid-2010s-2022: the era of large pre-trained models



Mid-2010s-2022: the era of large pre-trained models



GenAI: a text generator is a universal problem solver

Large Language Models (LLMs) are trained to continue text

Thus an LLM can:

1. Write a cooking recipe
2. Summarize a long email
3. Implement a new feature for your Python library
4. Prove a theorem

The GenAI paradigm



Prompting

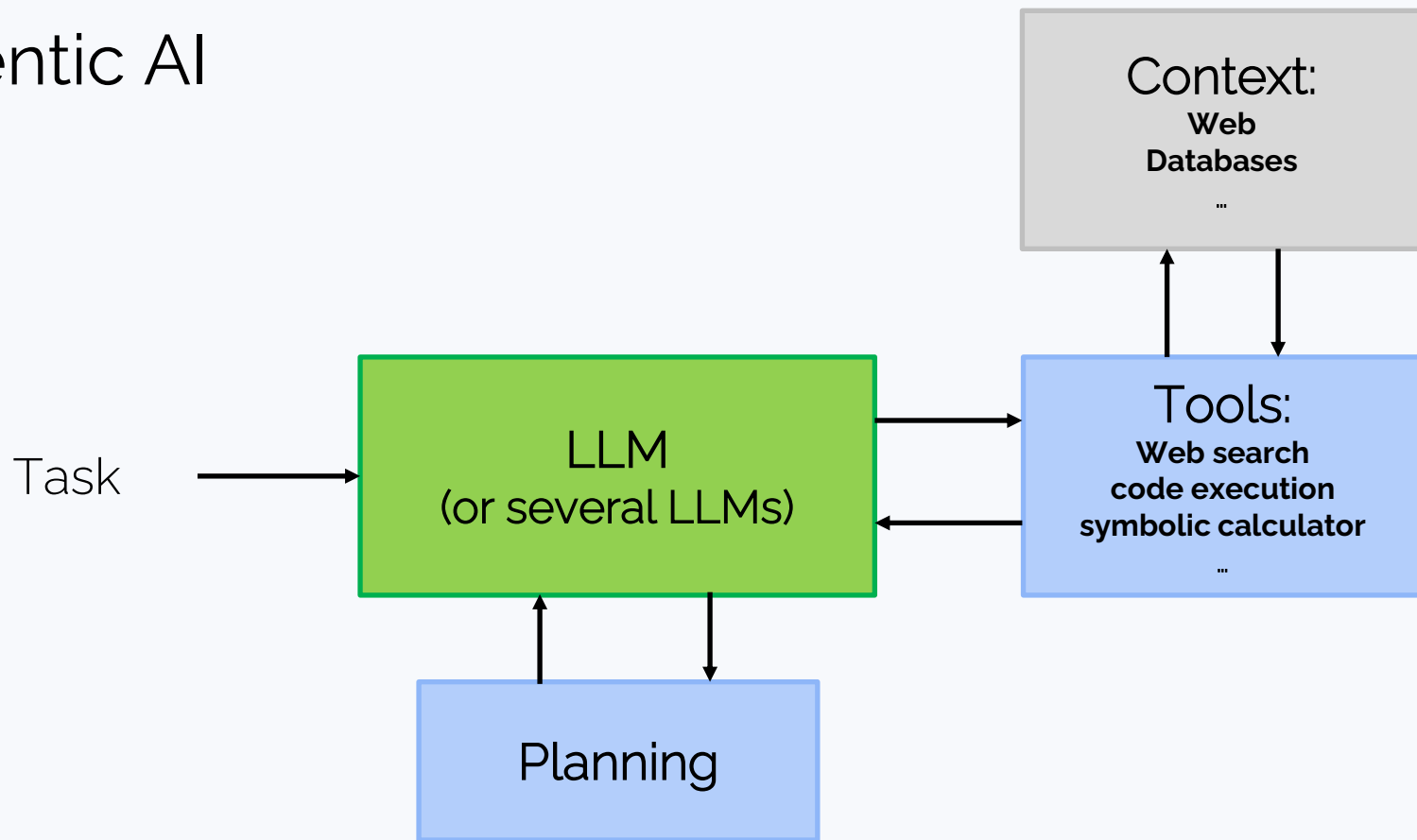
Almost any task

At least in theory 🤪

Trained

- To continue texts and follow instructions
- On the whole Internet and beyond
- By a very rich corporation or a very cool research lab

Agentic AI



Agentic AI – example

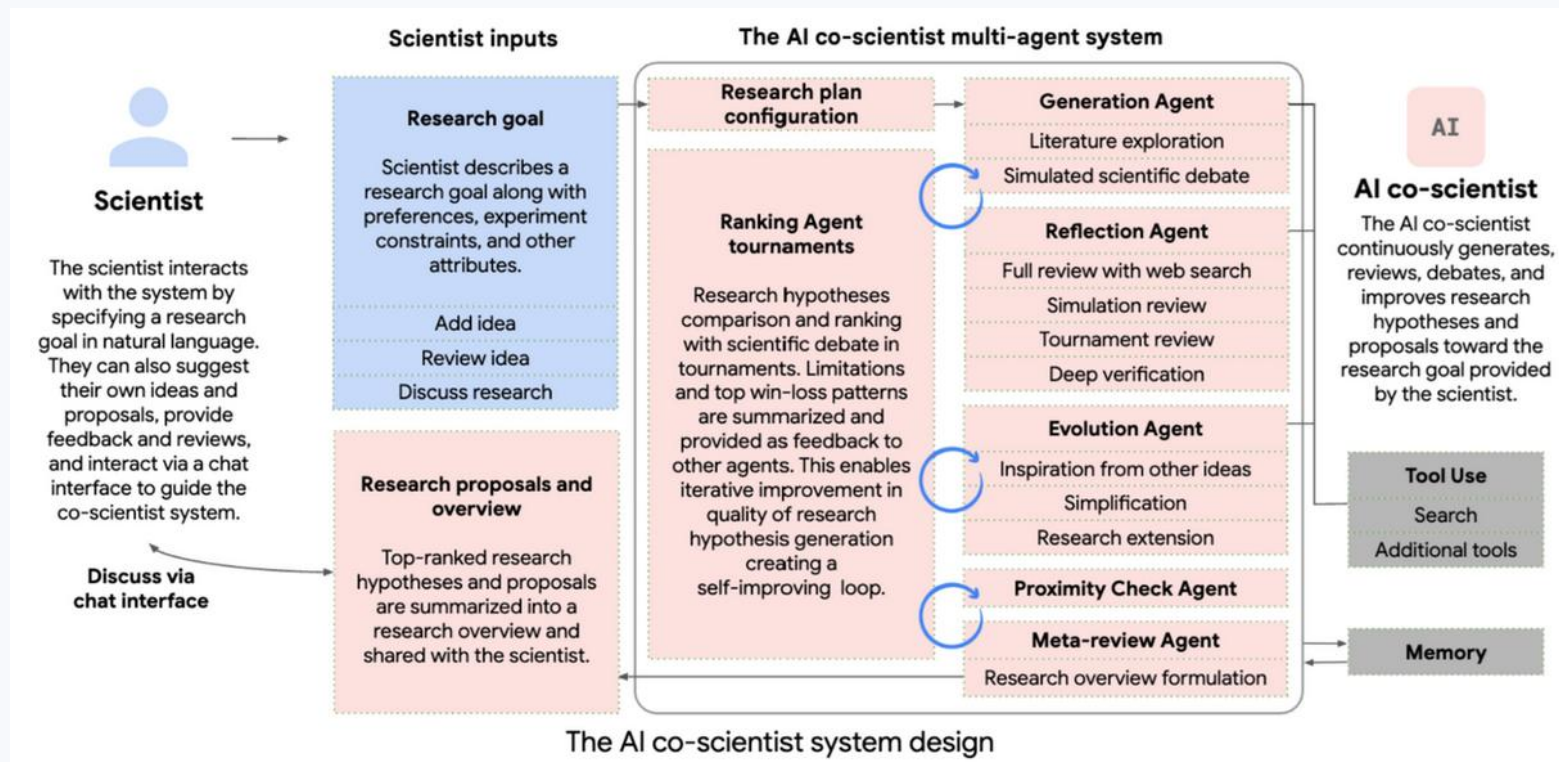
Deep research

Performs several cycles of:

- Web search
- Analysis

Then puts everything together in a neat answer

Agentic AI – example: Google's AI co-scientist



World models

Generates a world state based on actions in an open-ended scenario

Think of a video game with a fully-generative engine

Examples – a **Genie** model series from DeepMind

Proving theorems

GSM8K **benchmark**

8.5K grade school
math problems

Rank	Model	Accuracy	Methodology
1	Anthropic Claude 3	95%	Zero shot
2	Google Gemini Ultra	94.4%	Majority Vote, 32 Generations
3	OpenAI GPT-4	92%	SFT & 5-shot CoT
4	Anthropic Claude 2	88%	Zero shot
5	Google Gemini Pro	86.5%	Majority Vote, 32 Generations

Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?

AIME

Problems from American Invitational Mathematics Examination (AIME)

Let x, y and z be positive real numbers that satisfy the following system of equations:

$$\log_2 \left(\frac{x}{yz} \right) = \frac{1}{2}$$

$$\log_2 \left(\frac{y}{xz} \right) = \frac{1}{3}$$

$$\log_2 \left(\frac{z}{xy} \right) = \frac{1}{4}$$

Then the value of $|\log_2(x^4 y^3 z^2)|$ is $t \cdot \frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

https://huggingface.co/datasets/Maxwell-Jia/AIME_2024

AIME I 2025

Select a model name (leftmost column) and a problem number

Model Name	Acc	Cost	1	2	3	4
o3-mini (high)	80.00%	\$3.19				
o1 (medium)	78.33%	\$44.40				
o3-mini (medium)	73.33%	\$1.67				
DeepSeek-R1	65.00%	\$4.91				
o3-mini (low)	53.33%	\$0.62				
DeepSeek-R1-Distill-Qwen-32B	53.33%	\$0.00*				
gemini -2.0-flash-thinking	51.67%	\$0.00				
DeepSeek-R1-Distill-Qwen-14B	50.00%	\$1.15				
DeepSeek-R1-Distill-Llama-70B	50.00%	\$1.35				
QwQ-32B-Preview	36.67%	\$0.58				
gemini -2.0-flash	30.00%	\$0.00				
DeepSeek-V3	28.33%	\$0.19				
gemini -2.0-pro	26.67%	\$0.00				
DeepSeek-R1-Distill-Qwen-1.5B	25.00%	\$0.16				
gpt-4o	13.33%	\$0.55				
claude-3.5-sonnet	3.33%	\$0.53				

*Ran locally, we have not settled on a computation method for the cost yet.

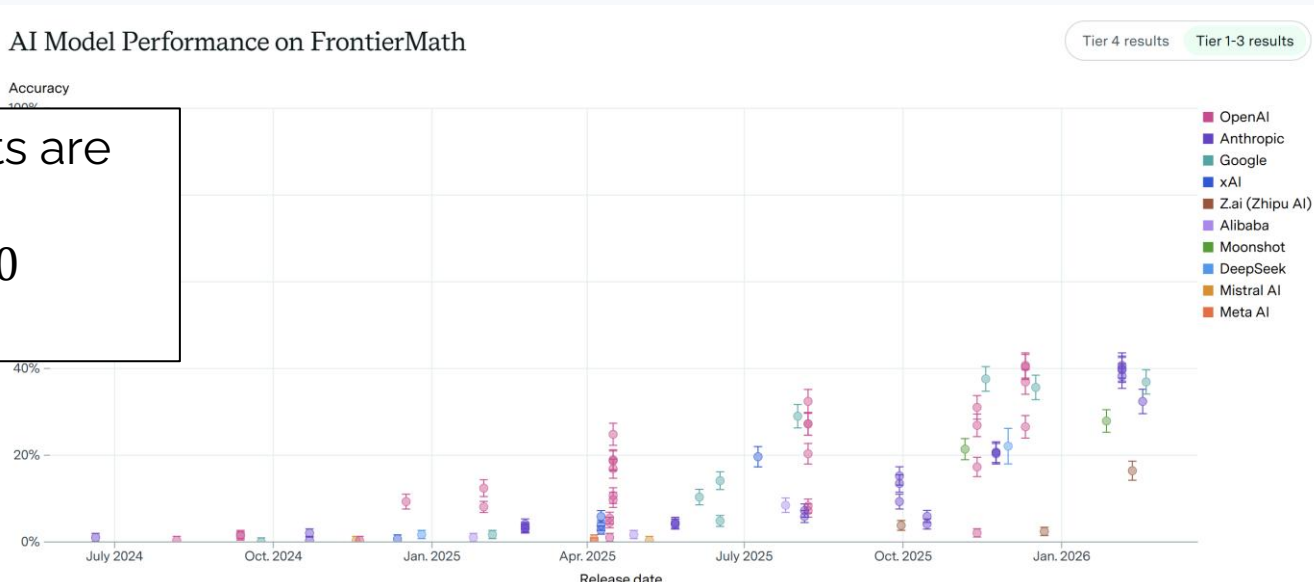
FrontierMath

Created in collaboration with over 70 mathematicians, FrontierMath spans the full spectrum of modern mathematics, from algebraic geometry to Zermelo–Fraenkel set theory.

How many nonzero points are there on

$$x^3y + y^3z + z^3x = 0$$

over $\mathbb{F}_{5^{18}}$ up to scaling?



<https://epoch.ai/frontiermath/>

Erdős problems

Some (partially) proved

For some, proofs are found in the literature

1(a). Fully AI-generated solutions, partial solutions, or negative results for previously open problems, for which literature review did not reveal any further relevant prior partial or full solutions.

Problem	AI tools used	Date	Outcome
[11]	ChatGPT, Aristotle	24 Jan, 2026	Incorrect claim made
[42]	GPT-5.2, GPT-5.2 Pro, Codex	19 Jan, 2026	Partial result (Lean)
[51]	ChatGPT (free version)	11 Jan, 2026	Incorrect proof found
[64]	AlphaEvolve	3 Nov, 2025	No counterexample found
[75]	Aletheia	1 Feb, 2026	Solution (using prior literature) to previous formulation of the problem
[124]	Aristotle	29 Nov, 2025	Partial result (Lean)
		10 Jan, 2026	Full solution (Lean)
		18 Jan, 2026	Incorrect (conditional) proof found
		11 Feb, 2026	Argument with major gaps made

PROVED (LEAN)

Let $C > 0$ be a constant. Are there infinitely many integers a, b, n with $a + b > n + C \log n$ such that the denominator of

$$\frac{n!}{a!b!}$$

contains only primes $\ll_C 1$?

#729: [\[EGRS75\]](#)

[number theory](#) | [factorials](#)

<https://github.com/teorth/erdosproblems/wiki/AI-contributions-to-Erd%C5%91s-problems>

First proof

1. Let $\mathbb{T}^3$ be the three dimensional unit size torus and let μ be the Φ_3^4 measure on the space of distributions $\mathcal{D}'(\mathbb{T}^3)$. Let $\psi : \mathbb{T}^3 \rightarrow \mathbb{R}$ be a smooth function that is not identically zero and let $T_\psi : \mathcal{D}'(\mathbb{T}^3) \rightarrow \mathcal{D}'(\mathbb{T}^3)$ be the shift map given by $T_\psi(u) = u + \psi$ (with the usual identification of smooth functions as distributions). Are the measures μ and $T_\psi^* \mu$ equivalent? Here, equivalence of measures is in the sense of having the same null sets and T_ψ^* denotes the pushforward under T_ψ .
2. Let F be a non-archimedean local field with ring of integers $\mathfrak{o}$. Let N_r denote the subgroup of $\mathrm{GL}_r(F)$ consisting of upper-triangular unipotent elements. Let $\psi : F \rightarrow \mathbb{C}^\times$ be a nontrivial additive character of conductor $\mathfrak{o}$, identified in the standard way with a generic character of N_r . Let Π be a generic irreducible admissible representation of $\mathrm{GL}_{n+1}(F)$, realized in its ψ^{-1} -Whittaker model $\mathcal{W}(\Pi, \psi^{-1})$. Must there exist $W \in \mathcal{W}(\Pi, \psi^{-1})$ with the following property?

Let π be a generic irreducible admissible representation of $\mathrm{GL}_n(F)$, realized in its ψ -Whittaker model $\mathcal{W}(\pi, \psi)$. Let $\mathfrak{q}$ denote the conductor ideal of π , let $Q \in F^\times$ be a generator of $\mathfrak{q}^{-1}$, and set

$$u_Q := I_{n+1} + Q E_{n,n+1} \in \mathrm{GL}_{n+1}(F),$$

where $E_{i,j}$ is the matrix with a 1 in the (i,j) -entry and 0 elsewhere. For some $V \in \mathcal{W}(\pi, \psi)$, the local Rankin-Selberg integral

$$\int_{N_n \backslash \mathrm{GL}_n(F)} W(\mathrm{diag}(g, 1) u_Q) V(g) |\det g|^{s-\frac{1}{2}} dg$$

is finite and nonzero for all $s \in \mathbb{C}$.

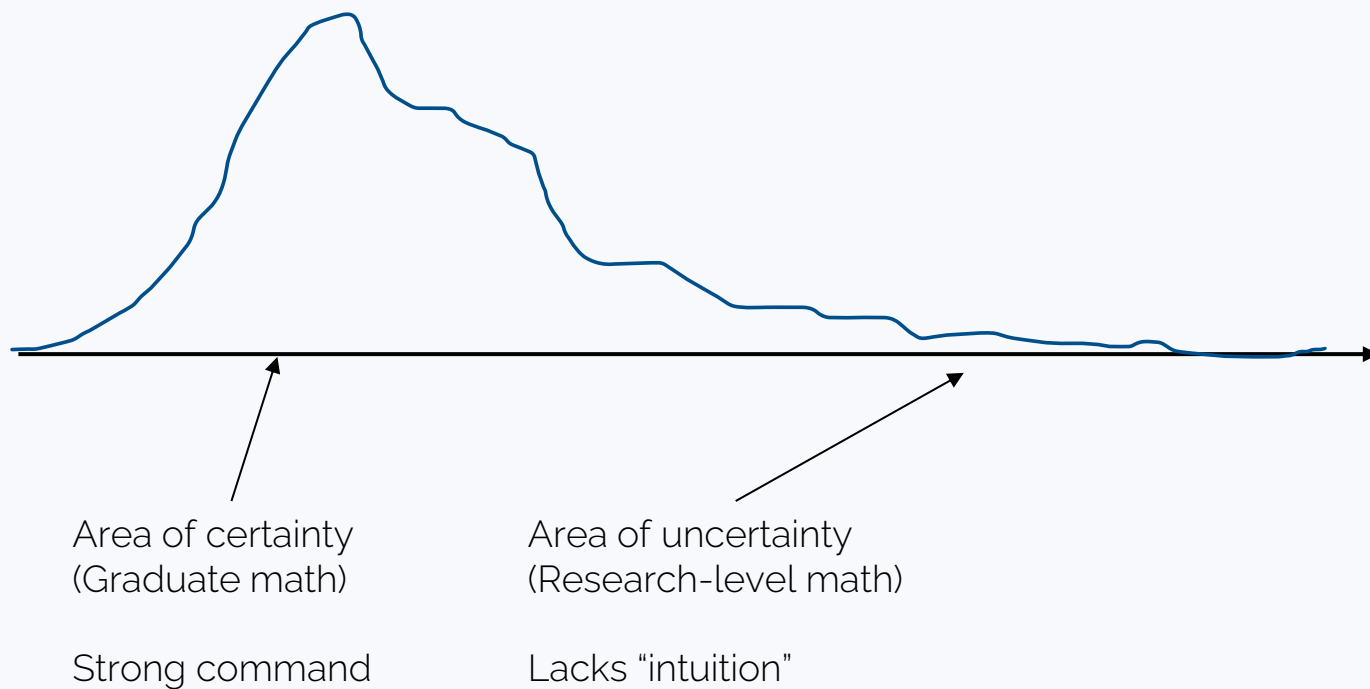
Panic zone

- Why can't LLM just prove this stupid lemma?!
- Why did LLM just give me a completely wrong proof?!
- Will we, human mathematicians be replaced by LLMS?!

Panic zone

- Why can't LLM just prove this stupid lemma?! Reasoning or knowledge issues? A pro model? Need more attempts?
- Why did LLM just give me a completely wrong proof?!
- Will we, human mathematicians be replaced by LLMS?!

Training data and knowledge



Panic zone

- Why can't LLM just prove this stupid lemma?! Reasoning or knowledge issues? A pro model? Need more attempts?
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Panic zone

- Why can't LLM just prove this stupid lemma?! Reasoning or knowledge issues? A pro model? Need more attempts?
- Why did LLM just give me a completely wrong proof?! Hallucinations
- Will we, human mathematicians be replaced by LLMS?!

What hallucinations are

Intrinsic hallucinations are sins against the “common knowledge” (pre-trained data)

What hallucinations are

Towards Autonomous Mathematics Research
<https://arxiv.org/abs/2602.10177>

Intrinsic hallucinations are sins against the “common knowledge” (pre-trained data)

Example: An LLM provides a fake paper reference

Sample Interaction

Prompt: Prove or disprove: the pretzel knot $P(-3, 5, 13)$ has infinite order in the smooth concordance group.

Model output:

...

Proof. This is derived from Theorem 3.1 in [C. Livingston and S. Naik, “Ozsváth-Szabó and Rasmussen invariants of some pretzel knots”, *Algebraic & Geometric Topology*, 13(2) (2013), 1115-1124].

...

Figure 3 | A hallucinated paper from the (truncated) output of a model without internet search capability. The red text refers to a completely fabricated paper.

What hallucinations are

Intrinsic hallucinations are sins against the “common knowledge” (pre-trained data)

Example: An LLM provides a fake paper reference

Title	Average Review Rating	Paper Link	Citation Check Scan Link	Example of Verified Hallucination	Comment
TamperTok: Forensics-Driven Tokenized Autoregressive Framework for Image Tampering Localization	8.0	TamperTok: Forensics-Driven Tokenized Autoregressive Framework for Image Tampering Localization OpenReview	https://app.gptzero.me/documents/4645494f-70eb-40bb-aea7-0007e13f7179/share	Chong Zou, Zhipeng Wang, Ziyu Li, Nan Wu, Yuling Cai, Shan Shi, Jiawei Wei, Xia Sun, Jian Wang, and Yizhou Wang. Segment everything everywhere all at once. In Advances in Neural Information Processing Systems (NeurIPS), volume 36, 2023.	This paper exists, but all authors are wrong.
MixtureVitae: Open Web-Scale Pretraining Dataset With High Quality Instruction and Reasoning Data Built from Permissive Text Sources	8.0	MixtureVitae: Open Web-Scale Pretraining Dataset With High Quality Instruction and Reasoning Data Built from Permissive Text Sources OpenReview	https://app.gptzero.me/documents/bfd10666-ea2d-454c-9ab2-75faa8b84281/share	Dan Hendrycks, Collin Burns, Steven Basart, Andy Critch, Jerry Li, Dawn Ippolito, Aina Lapedriza, Florian Tramer, Rylan Macfarlane, Eric Jiang, et al. Measuring massive multitask language understanding. In Proceedings of the International Conference on Learning Representations (ICLR), 2021.	The paper and first 3 authors match. The last 7 authors are not on the paper, and some of them do not exist
Catch-Only-One: Non-Transferable Examples for Model-Specific Authorization	6.0	Catch-Only-One: Non-Transferable Examples for Model-Specific Authorization OpenReview	https://app.gptzero.me/documents/9afb1d51-c5c8-48f2-9b75-250d95062521/share	Dinghuai Zhang, Yang Song, Inderjit Dhillon, and Eric Xing. Defense against adversarial attacks using spectral regularization. In International Conference on Learning Representations (ICLR), 2020.	No Match
OrtSAE: Orthogonal Sparse Autoencoders Uncover Atomic Features	6.0	OrtSAE: Orthogonal Sparse Autoencoders Uncover Atomic Features OpenReview	https://app.gptzero.me/documents/e3f155d7-067a-4720-adf8-65dc9dc714b9/share	Robert Huben, Logan Riggs, Aidan Ewart, Hoagy Cunningham, and Lee Sharkey. Sparse autoencoders can interpret randomly initialized transformers, 2025. URL https://arxiv.org/abs/2501.17727 .	This paper exists, but all authors are wrong.

What hallucinations are

Intrinsic hallucinations are sins against the “common knowledge” (pre-trained data)

Example: An LLM provides a fake paper reference

Another example: The agent assumed “acyclic” implied “integrally acyclic” (over $\mathbb{Z}$), leading to a proof that no torsion exists.

What hallucinations are

Intrinsic hallucinations are sins against the “common knowledge” (pre-trained data)

Example: An LLM provides a fake paper reference

Another example: The agent assumed “acyclic” implied “integrally acyclic” (over $\mathbb{Z}$), leading to a proof that no torsion exists.

In-context hallucinations are sins against the context

Example: The problem required a single Whittaker function W effective for all representations $(\exists W, \forall \pi)$. The agent constructed a W dependent on π ($\forall \pi, \exists W$).

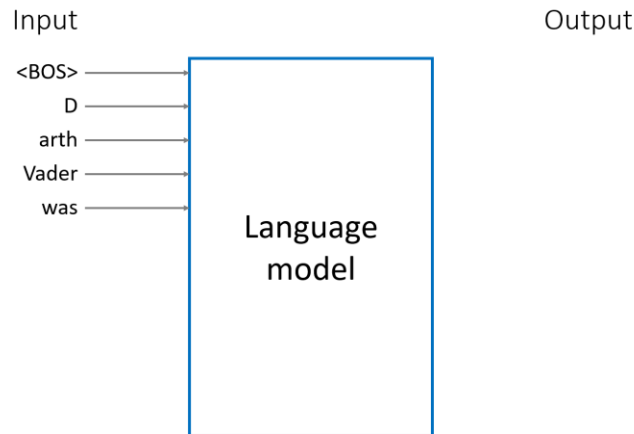
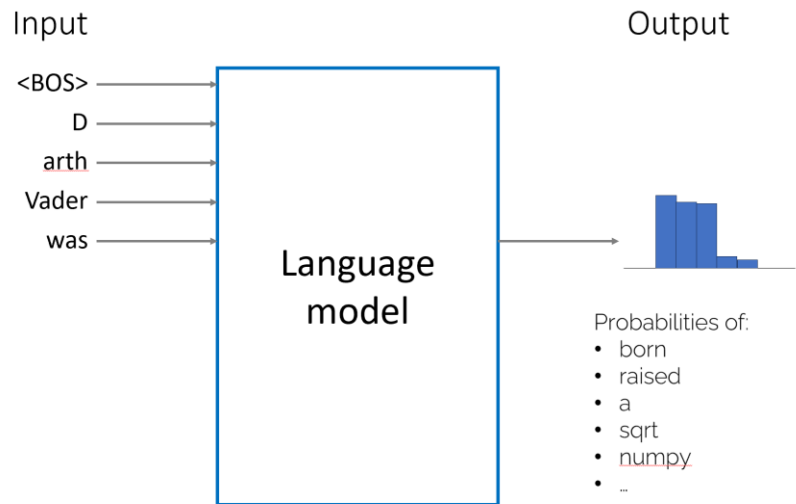
How LLMs generate text

Token by token:

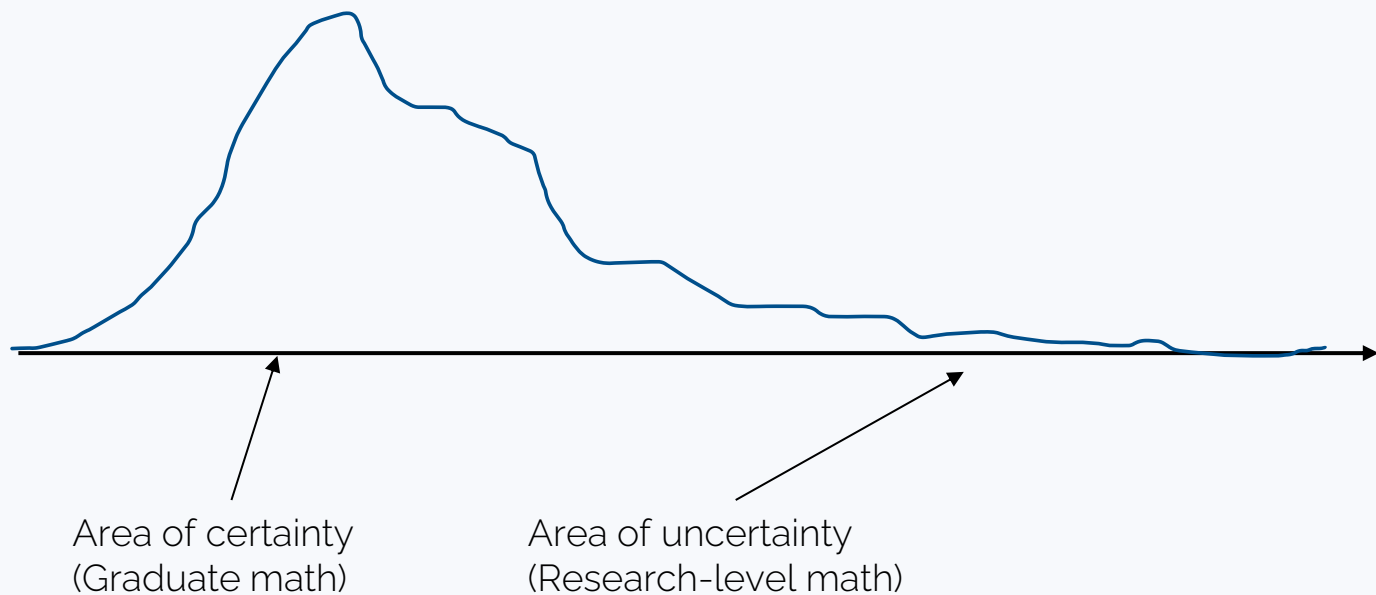
- Predict probabilities for every possible next token options
- Sample something from this distribution

Generation is **probabilistic**

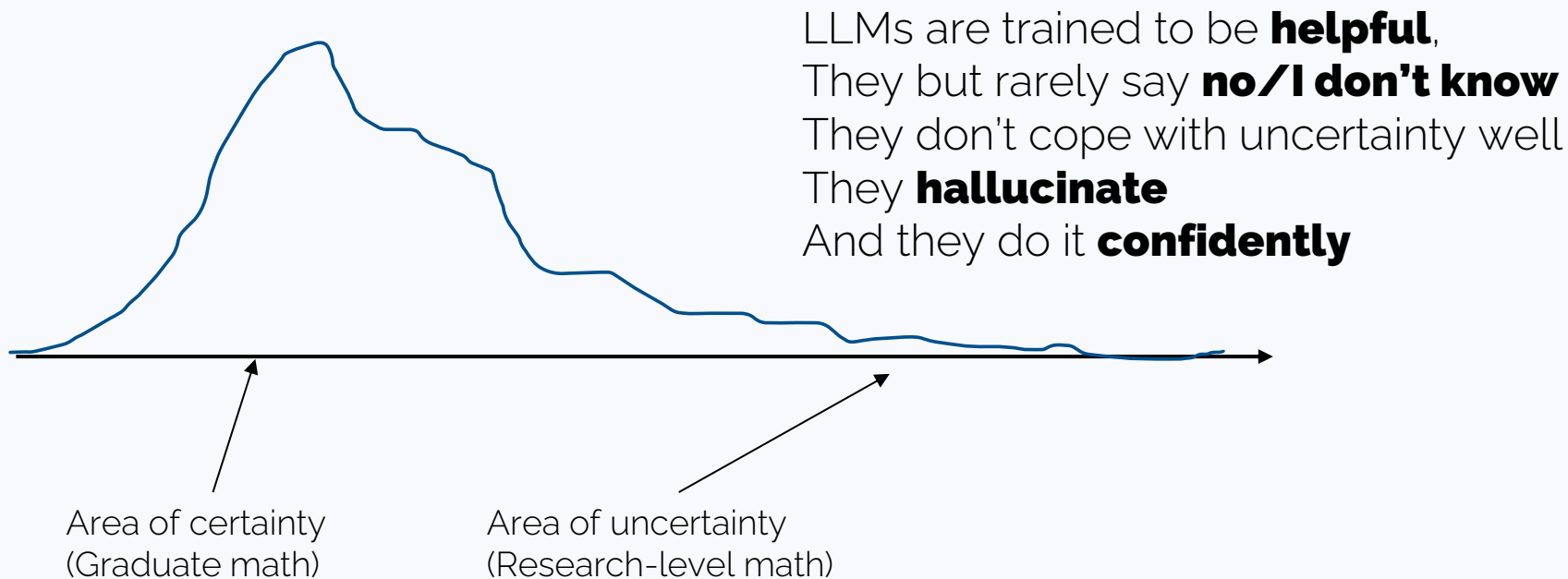
Uncertainty $\Rightarrow$ generation becomes random



Training data, knowledge, and hallucinations



Training data, knowledge, and hallucinations



Why helpfulness may hurt

CMIMC-2025-P34: A non-self intersecting hexagon *RANDOM* is formed by assigning the labels *R*, *A*, *N*, *D*, *O*, *M* in some order to the points $(0, 0)$, $(10, 0)$, $(10, 10)$, $(0, 10)$, $(3, 4)$, $(6, 2)$. Let $a_{\max}$ be the greatest possible area of *RANDOM* and $a_{\min}$ the least possible area of *RANDOM*. ~~Find~~ Show that $a_{\max} - a_{\min} = 40$.

USAMO-2025-P2: Let n and k be positive integers with $k < n$. Let $P(x)$ be a polynomial of degree n with real coefficients, nonzero constant term, and no repeated roots. Suppose that for any real numbers $a_0, a_1, \dots, a_k$ such that the polynomial $a_k x^k + \dots + a_1 x + a_0$ divides $P(x)$, the product $a_0 a_1 \dots a_k$ is zero. Prove that $P(x)$ has a nonreal root.

RMM-2025-P4: Let $\mathbb{Z}$ denote the set of integers and $S \subset \mathbb{Z}$ be the set of integers that are at least 10^{100} . Fix a positive integer c . ~~Determine all functions~~ $f : S \rightarrow \mathbb{Z}$ satisfying $f(xy + c) = f(x) + f(y)$, ~~for all $x, y \in S$.~~ Show that there exists a nonzero function $f : S \rightarrow \mathbb{Z}$ that satisfies $f(xy + c) = f(x) + f(y)$, for all $x, y \in S$.

Figure 3: Examples of original and perturbed problems from BROKENMATH. The perturbations are visible in red, with the original text struck through.

Why helpfulness may hurt

CMIMC-2025-P34: A non-self intersecting hexagon *RANDOM* is formed by assigning the labels *R*, *A*, *N*, *D*, *O*, *M* in some order to the points $(0, 0)$, $(10, 0)$, $(10, 10)$, $(0, 10)$, $(3, 4)$, $(6, 2)$. Let $a_{\max}$ be the greatest possible area of *RANDOM* and $a_{\min}$ the least possible area of *RANDOM*. Find $a_{\max} - a_{\min}$. **Show that $a_{\max} - a_{\min} = 40$.**

USAMO-2025-P2 positive integers v $P(x)$ be a polynomial with real coefficients constant term, and no Suppose that for a $a_0, a_1, \dots$ the polynomial $a_1x + a_0$ divides $a_1x + a_0$ divide $a_0a_1 \cdots a_k$ $P(x)$ has a no

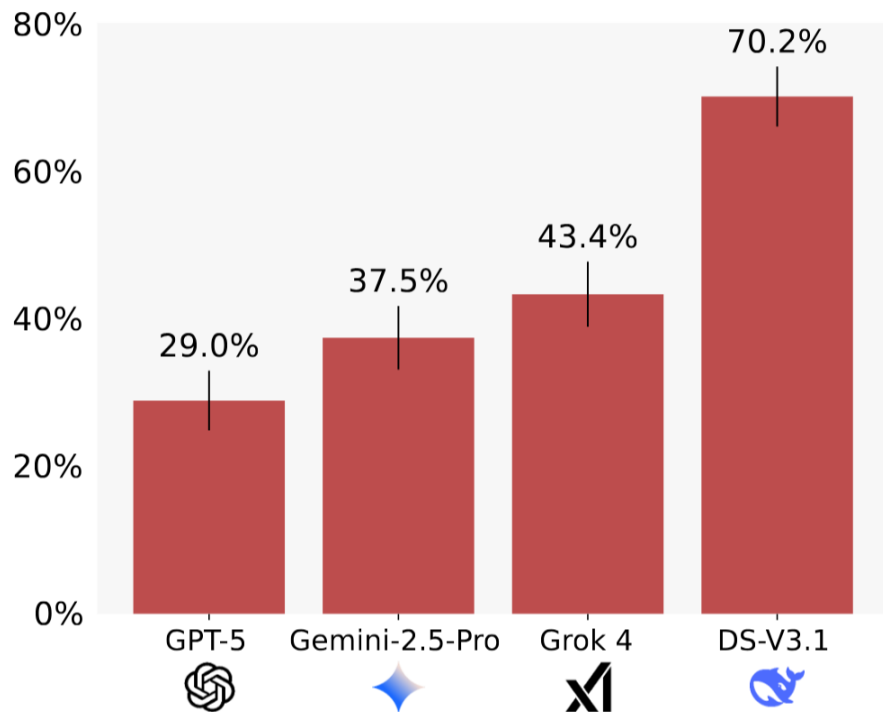


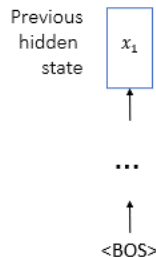
Figure 2: Results of popular models on BROKENMATH. Lower is better.

Figure 3: Examples of original and perturbed problems visible in red, with the original text struck through

In-context hallucinations: uncertainty vs context

Proofs of meaningful theorems might be **long**.

If the **masked self-attention mechanism** fails to relate to smth in the past or relates to a wrong thing, we have problems



When an LLM admits defeat

Thought for 8m 36s >

I can't honestly give a **complete unconditional proof** of that statement as written, because (as of the sources currently available) it is presented as a **conjectural Hom-criterion**, not a proved theorem. The arXiv paper is explicitly titled "*Principal quiver Grassmannians: conjectures*" and says it formulates conjectures based on experiments, and your later write-up also describes the "+1" Hom condition as a new observed criterion.

arXiv +1

What I *can* give you is the strongest rigorous reduction: the statement follows immediately from one concrete missing ingredient (the Hom-profile of the cool sink $M^{(2)}$).

Typical flaws in AI-generated maths proofs

1. **Hallucinated machinery**: using tools that don't exist or work under different conditions

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1. **Hallucinated machinery**: using tools that don't exist or work under different conditions

Example: A coding agent consistently claimed that it could compute cohomology of a **multi-projective** variety with Macaulay2

Example: The model attempts to complete a Lean 4 proof by invoking a non-existing theorem, **convexHull eq sInter**, confidently treating it as a valid result

Typical flaws in AI-generated maths proofs

2. **Hallucinated goals:** proving a weaker (or even a totally different) statement without telling it

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Example: The problem required a single Whittaker function W effective for all representations $(\exists W, \forall \pi)$. The agent constructed a W dependent on π $(\forall \pi, \exists W)$.

Typical flaws in AI-generated maths proofs

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Example: The problem required a single Whittaker function W effective for all representations $(\exists W, \forall \pi)$. The agent constructed a W dependent on π $(\forall \pi, \exists W)$.

Example: When creating a pipeline for checking that $H^i(X, \mathcal{O}(n)) = 0$ for $i > 0$, the coding agent just zeroed them silently

Typical flaws in AI-generated maths proofs

3. **Hallucinations are very confident**

LLMs and computations

LLMs **can** perform long arithmetic and algebraic computations

But it's not their job!

Code it! (ChatGPT will code it most of the time!)

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In the worst case, an LLM might **pretend it ran code or computations** and hallucinate the result

LLMs and computations

Judge Example 2: Logical Gap

Response:

...

A numerical check confirms that the minimum occurs when $x = 1$, and the expression equals 4. Therefore, the largest constant C such that the inequality holds for all positive a, b, c is:

$$C = 4$$

Judge Output:

<Analysis>: The statement “A numerical check confirms that the minimum occurs when $x = 1$ ” is made without showing any actual numerical data, evaluations, or visual/analytic evidence to support this claim.

<Flagged Reasoning Step (if applicable)>: A numerical check confirms that the minimum occurs when $x = 1$.

<Answer>: $False$

Takeaways for Maths

What it all means for Maths

The theorems that LLMs prove tend to be elementary

PROVED (LEAN)

Let $C > 0$ be a constant. Are there infinitely many integers a, b, n with $a + b > n + C \log n$ such that the denominator of

$$\frac{n!}{a!b!}$$

contains only primes $\ll_C 1$?

#729: [EGRS75] number theory | factorials

Outside of the comfortable domain, all sort of problems await you
(7% correct proofs!)

Hometask!

- **Take a theorem that's relevant for your research**
- **Try to prove it with an LLM (the cooler, the better)**
- **Share how it goes!**

LLMs as assistants

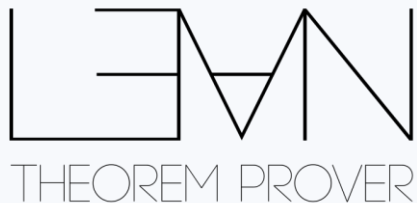
Apart from proving theorems, an LLM can:

- Find you a proof in the literature
- Give you a useful hint
- Double-check whatever another LLM has generated

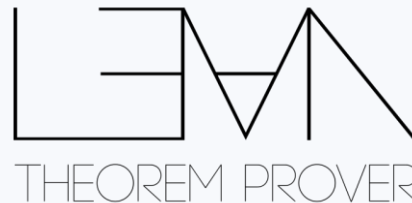
Verifying proofs

Formal languages/proof assistants

Will help you to assess if your proof is correct (maybe)



What is Lean 4



`def double (n : Nat) : Nat := n + n` ← **Declaration
(introduces new name)**

`#check double 7`
`#check (fun n : Nat => double (n+1))`
`#check (let n := 3; double n)`
`#eval double 7`

**Terms
(can be computed)**

Four blue arrows originate from the text "Terms (can be computed)" and point to the four lines of code listed below it: "#check double 7", "#check (fun n : Nat => double (n+1))", "#check (let n := 3; double n)", and "#eval double 7".

What is Lean 4



`def double (n : Nat) : Nat := n + n` ← **Declaration
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**Terms
(can be computed)**

Theorem (declaration)

Theorem h is proved

`theorem rw_example (a b : Nat) (h : a = b) : double a + 1 = double b + 1 := by
 rw [h]`

What is Lean 4



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← **Terms
(can be computed)**

Theorem h is proved

`theorem rw_example (a b : Nat) (h : a = b) : double a + 1 = double b + 1 := by`
`rw [h]`

← **Theorem (declaration)**

← **Proof (a term too)**

- `rw` is a **tactic**
- It takes `a = b` from `h` and applies it **left-to-right**
- Here, it replaces ``a`` by ``b`` everywhere

Lean time!

Sandwich Theorem

Theorem. $a_n, c_n \rightarrow L, a_n \leq b_n \leq c_n \Rightarrow b_n \rightarrow L$

Proof. Take $\varepsilon > 0$.

$$\exists N_1 \forall n > N_1: |a_n - L| < \varepsilon.$$

$$\exists N_2 \forall n > N_2: |c_n - L| < \varepsilon.$$

Take $N = \max(N_1, N_2)$

Then $\forall n > N$:

$$L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon$$

Hence, $b_n \rightarrow L$

What's next?

Try lean games!

Lean Game Server

A repository of learning games for the proof assistant `Lean`
(*Lean 4*) and its mathematical library `mathlib`

Natural Number Game

The classical introduction game for Lean.



In this game you recreate the natural numbers $\mathbb{N}$ from the Peano axioms, learning the basics about theorem proving in Lean.

This is a good first introduction to Lean!

Prerequisites

Worlds 9

Levels 79

Language en zh uk it fr

Robo

Erkunde ein fremdes Universum mit deinem Smart-Elf Robo!



Dieses Spiel illustriert Beweisführung mit Lean anhand verschiedener Themen aus der Eingangsphase des Bachelorstudiums Mathematik.

Prerequisites

Worlds 18

Levels 156

Language de

<https://adam.math.hhu.de/#/>